

Multiple Choice

1. **Answer:** A
Options: A) gamma rays; B) microwaves; C) ultraviolet radiation; D) visible light
Note: Correct option: A - gamma rays
2. **Answer:** D
Options: A) 2; B) 4; C) 6; D) 10
Note: Correct option: D - 10
3. **Answer:** D
Options: A) 0.25 W; B) 0.5 W; C) 1 W; D) 4 W
Note: Correct option: D - 4 W
4. **Answer:** D
Options: A) Bosons have symmetric wave functions and obey the Pauli exclusion principle.; B) Bosons have antisymmetric wave functions and do not obey the Pauli exclusion principle.; C) Fermions have symmetric wave functions and obey the Pauli exclusion principle.; D) Fermions have antisymmetric wave functions and obey the Pauli exclusion principle.
Note: Correct option: D - Fermions have antisymmetric wave functions and obey the Pauli exclusion principle.
5. **Answer:** D
Options: A) $0.1 \text{ GeV}/c^2$; B) $0.2 \text{ GeV}/c^2$; C) $0.5 \text{ GeV}/c^2$; D) $1.0 \text{ GeV}/c^2$
Note: Correct option: D - $1.0 \text{ GeV}/c^2$

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'a dye laser'.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 25%. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 1/3. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key